proof: $\begin{pmatrix} f(p+k,1) \\ f(p+k,2) \\ \vdots \\ f(p+k,d-1) \\ f(p+k,d) \end{pmatrix} = \begin{pmatrix} R(w_1k) & & & \\ & R(w_3k) & & \\ & & \ddots & \\ & & & R(w_{d-1}k) \end{pmatrix} \begin{pmatrix} f(p,1) \\ f(p,2) \\ \vdots \\ f(p,d-1) \\ f(p,d) \end{pmatrix}$

$\therefore f(p+k) = R(k) f(p)$

3. $\begin{pmatrix} s(a+b) \\ c(a+b) \end{pmatrix} = \begin{pmatrix} sacb + casb \\ cacb - sasb \end{pmatrix}$ (from 02187)

3.1 $= \begin{pmatrix} cb & sb \\ -sb & cb \end{pmatrix} \begin{pmatrix} sa \\ ca \end{pmatrix}$

3.2 $\begin{pmatrix} sa \\ ca \end{pmatrix}^T \begin{pmatrix} sb \\ cb \end{pmatrix} = sasb + cacb$

$= \cos(a-b)$

(depend only on the difference $a-b$)

4. $\begin{pmatrix} f(p+k,j) \\ f(p+k,j+1) \end{pmatrix} = R(w_jk) \begin{pmatrix} f(p,j) \\ f(p,j+1) \end{pmatrix}$, j odd.

here $R(w_jk) = \begin{pmatrix} cw_jk & sw_jk \\ -sw_jk & cw_jk \end{pmatrix}$

proof: left $= \begin{pmatrix} \sin(w_j(p+k-1)) \\ \cos(w_j(p+k-1)) \end{pmatrix}$

$= \begin{pmatrix} s(a+b) \\ c(a+b) \end{pmatrix}$, $a = w_j(p-1)$, $b = w_jk$

$= R(b) \begin{pmatrix} sa \\ ca \end{pmatrix}$ (from 3.1)

$=$ right

5. d even $\Rightarrow f(p+k) = R(k) f(p)$

here $R(k) = \text{diag}(R(w_1k), R(w_3k), \cdots, R(w_{d-1}k))$

$R(w_jk) = \begin{pmatrix} cw_jk & sw_jk \\ -sw_jk & cw_jk \end{pmatrix}$, j odd.

6. d odd, solve $f(p+k)$

solve: ① (method 1)

$\because \begin{pmatrix} f(p+k) \\ f(p+k,d+1) \end{pmatrix} = R_1(k) \begin{pmatrix} f(p) \\ f(p,d+1) \end{pmatrix}$

$f(p,d+1) = \cos(w_d(p-1))$

$R_1(k) = \text{diag}(R(w_1k), R(w_3k), \cdots, R(w_dk))$

$\Rightarrow f(p+k)$

② (method 2)

$f(p+k,1:d-1) = R_2(k) f(p,1:d-1)$

$R_2(k) = \text{diag}(R(w_1k), R(w_3k), \cdots, R(w_{d-2}k))$

$f(p+k,d) = \sin(w_d(p+k-1))$

7. $f(p)^T f(q) = \begin{pmatrix} s(w_1(p-1)) \\ c(w_1(p-1)) \\ s(w_3(p-1)) \\ \cdots \end{pmatrix}^T \begin{pmatrix} s(w_1(q-1)) \\ c(w_1(q-1)) \\ s(w_3(q-1)) \\ \cdots \end{pmatrix}$

$= \sum_{i=1}^{\lfloor \frac{d}{2} \rfloor} \cos(w_{2i-1}(p-q)) + r$

here $r = \begin{cases} \sin(w_d(p-1))\sin(w_d(q-1)), & \text{if } d \text{ odd} \\ 0, & \text{otherwise.} \end{cases}$

Sinusoidal Positional Encoding

1. $f(p)$, PE function.

1.2 $f: \{1, \cdots, L\} \to R^d$, $f(p) = \begin{pmatrix} f(p,1) \\ f(p,2) \\ \vdots \\ f(p,d) \end{pmatrix}$

here $f(p,j) := \begin{cases} \sin(w_j(p-1)), & j \text{ odd} \\ \cos(w_{j-1}(p-1)), & j \text{ even} \end{cases}$

$w_j := \frac{1}{B^{(j-1)/d}}$, j odd, $B = 10000$

$p = 1, \cdots, L$. $j = 1, \cdots, d$.

1.3 $F = \begin{pmatrix} f(1)^T \\ \vdots \\ f(L)^T \end{pmatrix}$ positional-encoding matrix $\in R^{L \times d}$

$F_{p,j} = f(p,j)$.

2. graph of $f(p,j)$

2.2 for w_j, period $T_j = \frac{2\pi}{w_j} = 2\pi B^{\frac{j-1}{d}}$

$\because j$ odd, $j \le d$ $\therefore 0 \le \frac{j-1}{d} < 1$

$\therefore 1 \le B^{(j-1)/d} < B$ $\therefore 2\pi \le T_j < B \cdot 2\pi$

$\therefore$ sketch of $f(p,j)$:

